

HAT-P-3b

R-0999

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-3_240409 · created 2026-07-30T01:06:46+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460409.7027 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.145 +/- 0.043
Transit depth (Rp/Rs)^2	2.09 +/- 1.24 [%]
Orbital Inclination (inc)	89.75 +/- 2.28
Airmass coefficient 1 (a1)	0.57 +/- 0.25
Airmass coefficient 2 (a2)	0.28 +/- 0.4
Scatter in the residuals of the lightcurve fit is	37.3 %
Best Comparison Star	#3 - [416, 123]
Optimal Aperture	2.2
Optimal Annulus	5.7
Transit Duration (day)	0.094 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.145 ± 0.043 vs 0.1063 (deviation 0.9σ)
Duration fit vs published	0.094 d vs 0.0867 d (deviation 0.56σ)
Mid-time O–C	3.4 min
Depth SNR	3.4
Residual scatter	37.3 %

Light curve

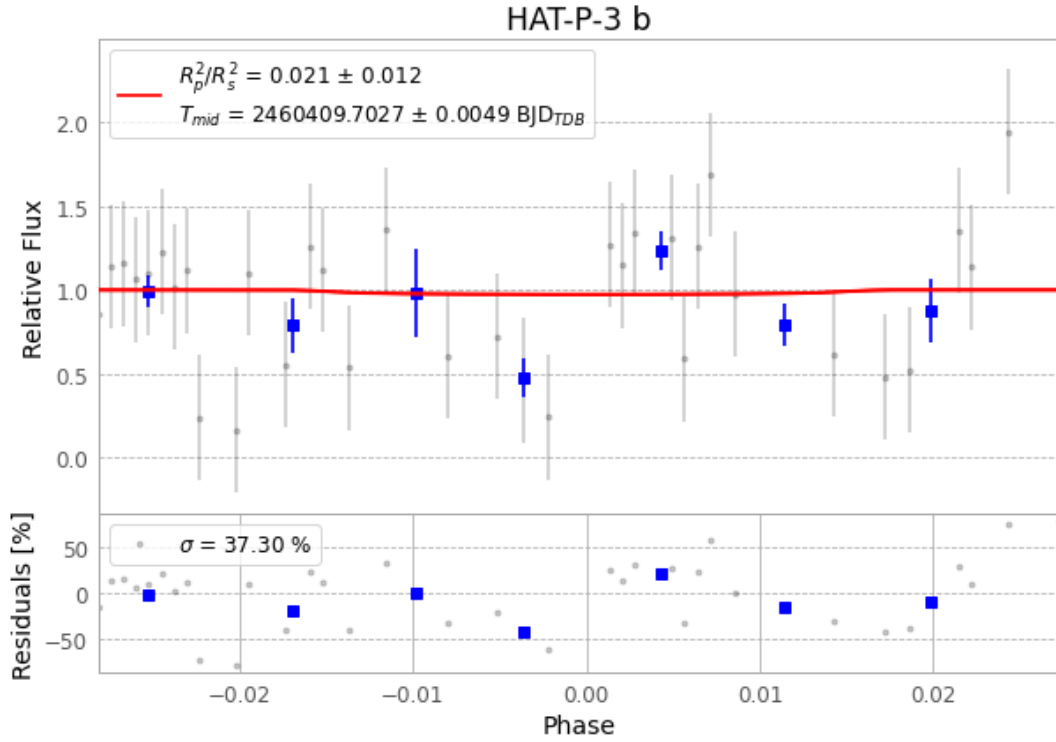


Figure 1 — FinalLightCurve_HAT-P-3 b_2024-04-08.png (SHA-256 63730959084fd1ad...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [340, 244], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1ec267f3006f9ff4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

80 FITS frames (53 MB), HATP-3240409024815.FITS → HATP-3240409064515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-3-240409.log (db45d0b540f5...), FinalLightCurve_HAT-P-3 b_2024-04-08.png (63730959084f...), FinalLightCurve_HAT-P-3 b_2024-04-08.csv (ccac6d17bb89...)

Compute resources

Wall time 195.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 01:06Z.